Types of Radiation

Exam code: 0625 & 0972

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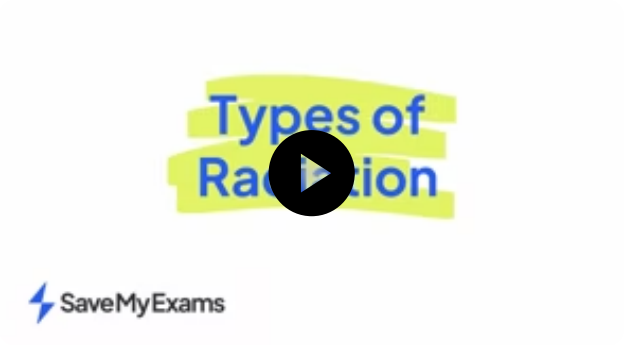
Updated on 8 April 2025



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Radioactive decay

- The emission of radiation from a nucleus is **spontaneous** and **random** in direction
- This random process of radioactive decay means:
  - There is an **equal probability** of any nucleus decaying
  - It cannot be known **which particular nucleus will decay next**
  - It cannot be known **at what time a particular nucleus will decay**
  - The rate of decay is **unaffected** by the surrounding conditions
  - It is only possible to estimate the **probability** of a nuclei decaying in a given time period

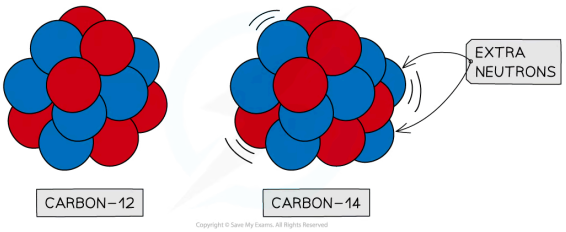
Unstable nuclei

- Some atomic nuclei are **unstable**
- This is because of an imbalance in the forces within the nucleus
  - Forces exist between the particles in the nucleus
- Instability is commonly due to:
  - The nucleus having **too many** protons or neutrons
  - The nucleus being **very large**
- An example of an **unstable** nucleus is carbon-14
  - This is an **isotope** of carbon

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

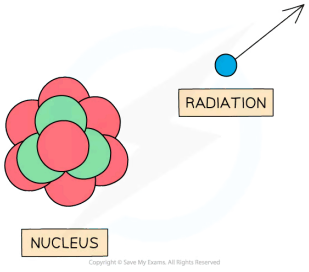
Isotopes of carbon



Carbon-12 is stable, whereas carbon-14 is unstable. This is because carbon-14 has two extra neutrons

- Unstable nuclei can **emit radiation** to become more stable
  - Radiation can be in the form of a **high-energy** particle or wave

Unstable nuclei emit radiation



Unstable nuclei decay by emitting high-energy particles or waves

- As the radiation moves away from the nucleus, it takes some energy with it
  - This reduces the overall energy of the nucleus
  - This makes the nucleus more **stable**

Worked Example

Which of the following statements is **not** true?

A Isotopes can be unstable because they have too many or too few neutrons

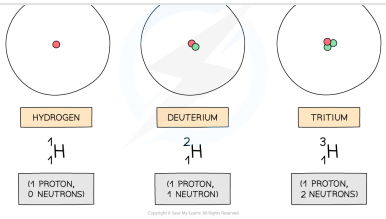
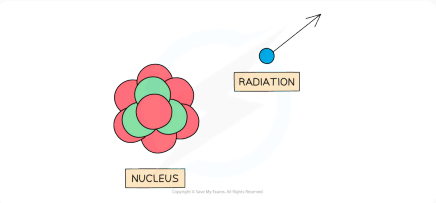
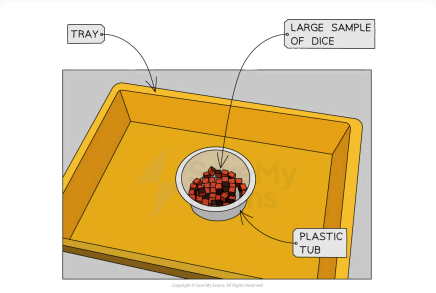
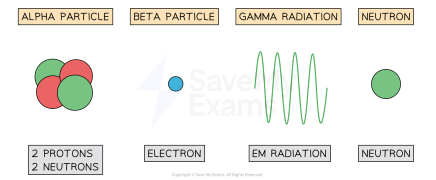
B The process of emitting particles or waves of energy from an unstable nucleus is called radioactive decay

C Scientists can predict when a nucleus will decay

D Radiation refers to the particles or waves emitted from a decaying nucleus

Answer: C

- Consider what you know about the statements above:

	
Radioactive decay	
Nuclei decay prediction	
Radiation	

- Answer A is **true.** The number of neutrons in a nucleus determines the stability
- Answer B is **true.** This is a suitable description of radioactive decay
- Answer D is **true.** Radiation is about emissions. It is different to radioactive particles
- Answer C is **not true**
- Radioactive decay is a random process
- It is not possible to predict precisely when a particular nucleus will decay

Examiner Tips and Tricks

The terms **unstable**, **random** and **decay** have very particular meanings in this topic. Remember to use them correctly when answering questions!

Types of radioactive decay

- Radioactive decay is a change in an unstable nucleus that can result in the emission of one of the following types of

    - **Gamma (γ)** radiation
- Remember that these changes are **spontaneous** and **random**

Alpha particles

- The radiation symbol for alpha is **α**
- An alpha particle is the same as a helium nucleus
    - This is because they consist of two neutrons and two protons
- Alpha particles have a charge of +2
    - This means they can be affected by an electric field

Beta particles

- The radiation symbol for beta is **β^-**
- Beta particles are fast-moving electrons
- They are produced in nuclei when a neutron changes into a proton and an electron
- Beta particles have a charge of -1
    - This means they can be affected by an electric field

Gamma rays

- The radiation symbol for gamma is **γ**
- Gamma rays are electromagnetic waves
- They have the highest energy of the different types of electromagnetic waves
- Gamma rays have no charge

Types of radioactive decay

Alpha particles, beta particles and gamma waves can be emitted from unstable nuclei

Alpha, beta & gamma emission

- α, β and γ radiation can be identified by the emission from a nucleus by recalling their:
    - Nature (what type of particle or radiation they are)
    - Their relative ionising effects (how easily they **ionise** other atoms)

table which shows the following trends down the table:
    - The range increases
    - Penetrating power increases
    - Ionisation decreases

Summary of the properties of nuclear radiation

Particle	Nature	Range in air	Penetrating power	Ionisin ability
Alpha (α)	helium nucleus (2 protons, 2 neutrons)	a few cm	low; stopped by a thin sheet of paper	high
Beta (β)	high-energy electron	a few 10s of cm	moderate; stopped by a few mm of aluminium foil or Perspex	modera
Gamma (γ)	electromagnetic wave	infinite	high; reduced by a few cm of lead	low

Penetrating power

- Alpha, beta and gamma radiation have different properties
- So they **penetrate** materials in different ways
    - This means they are each **stopped** by different materials

Penetrating power of alpha, beta and gamma radiation

Alpha, beta and gamma are different in how they penetrate materials. Alpha is the least penetrating, and gamma is the most penetrating

    - Gamma can pass through **aluminium**
- Gamma rays are only partially stopped by thick **lead**

Examiner Tips and Tricks

It is important to note that **beta particles** are only **stopped** by **aluminium** if it is a **few mm** thick. They **can** pass though aluminium which is **thinner** than this. This concept often comes up in exam questions.

Worked Example

A student has an unknown radioactive source and is trying to determine which type of radiation it emits. Using a Geiger-Muller tube, they measure the count rate when the source is placed behind different materials.

Their results are shown in the table below:

	No material between source and detector	**Paper between source and detector**	**5 mm aluminium between source and detector**	**5 mm lead between source and detector**
Count rate	4320	4218	256	34

Which type of radiation is being given off by the source?

A Alpha particles

B Beta particles

C Gamma rays

D Neutrons

Answer: B

- Consider the diagram showing penetrating power from above
- The answer is **not A** because the radiation passed through the paper almost unchanged
    - This means it is **not** alpha
- The answer is **not C or D** because the aluminium decreased the count rate significantly
    - This means it is **not** gamma (gamma penetrates aluminium)
    - This also means it is **not** neutrons (neutrons penetrate aluminium, however, you do not need to know this for your exam)

Examiner Tips and Tricks

Remembering the type of particle, penetration and ionising power for alpha, beta and gamma radiation is very important for your exam! Often the exam question will give some clues and you will have to choose which type of radiation it could be based on these.

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